BaiZe: From-Scratch Vision Encoder and Multimodal Grounding for Visual EDA Tasks

Abstract

Agentic assistants for electronic design automation (EDA) are already deployed in-house, but a correctness-critical slice of the flow—custom and analog design—resists them, because much of the correctness signal lives in *visual* artifacts: layout, schematic, and waveform views that a text-only model cannot observe.

We present the multimodal extension of **BaiZe**, an EDA agent built on the BaiZe language model [7]: a vision encoder pre-trained from scratch, aligned to the language backbone through a lightweight projector, and adapted to the semiconductor domain via reinforcement learning.

A larger ViT-L/14 encoder reaches 68.2% linear-probe on an internal protocol ($+3.9$ pp over ViT-B/16), yet the aligned MLLM averages only 26.1% over 11 multimodal benchmarks, indistinguishable from the smaller tower: encoder-quality gains do not transfer downstream at the 224-px single-tile setting.

We localise this gap and evaluate the full system on three tiers—12 general multimodal benchmarks, 4 domain (EDA) benchmarks, and the multimodal subset of an EDA flow benchmark—measuring whether a 196-token-per-image budget suffices for detail-dense views, and whether domain reinforcement learning closes the transfer gap.

We release the full recipe and an evaluation protocol. Headline measurements: the ViT-L/14 encoder reaches 68.2% linear-probe ($+3.9$ pp over ViT-B/16); the aligned MLLM averages 26.1% over 11 multimodal benchmarks (indistinguishable between towers); and a CPU-only domain-RL pilot on waveform understanding yields one controlled, significant gain (type classification $16\% \rightarrow 37\% \rightarrow 53\%$, p=0.0009, cheat rate 0) alongside honestly reported non-significant iterations, plus an audit bounding the ceiling of an image channel on the current EDA-flow benchmark.

1 Introduction

Agentic assistants for electronic design automation (EDA) are already deployed in-house, coordinating design flows through Flash-tier masters. Yet a correctness-critical slice of the flow resists them: in custom and analog design, the signal that determines whether a layout "works" lives in *visual* artifacts—layout, schematic, and waveform views—that a text-only model cannot observe. A subagent that can only read scripts and logs is blind to the very object it is designing, and must rely on an engineer to translate geometry into text—a bottleneck that scales with design volume.

Contributions. We build the multimodal core of **BaiZe**—an EDA agent—on top of the BaiZe language backbone [7], and ask a single question: *can a from-scratch vision encoder make such a small model see the visual EDA artifacts that carry the correctness signal?*

- **A from-scratch vision encoder.** BaiZe pre-trains its own vision encoder rather than reusing an off-the-shelf one, and selects the tower from iso-parameter candidates at the 500–600 M

scale. A ViT-L/14 tower reaches 68.2% linear-probe on an internal protocol, $+3.9$ pp above ViT-B/16, at a 196-token-per-image budget.

- **Token-bounded multimodal alignment.** BaiZe connects the encoder to the frozen language backbone through a lightweight projector, following the staged curriculum of open multimodal assistants, and asks whether a 196-token-per-image budget suffices for detail-dense layout and schematic views. We show that encoder-quality gains do *not* transfer downstream: the larger tower averages 26.1% over 11 benchmarks, indistinguishable from the smaller one, and we localise the gap to the 224-px single-tile setting and the alignment architecture.
- **Domain adaptation.** A final reinforcement-learning stage adapts the multimodal model to proprietary semiconductor practice. We instantiate it as a CPU-only pilot on *waveform understanding*—a verifiable probe skill where vision matters most—with pre-registered, falsifiable criteria (control arm, McNemar, bootstrap CI, cheat rate), and report one significant gain together with honestly documented negative iterations and an audit of where an image channel would actually help on a real EDA-flow benchmark.
- **Three-tier evaluation.** We evaluate end-to-end on 12 general multimodal benchmarks, 4 domain (EDA) benchmarks, and the multimodal subset of an EDA flow benchmark, under a single frozen execution-based protocol.

2 Background and Related Work

2.1 Multimodal Vision-Language Models

Vision-language models connect a vision encoder to a language model through a lightweight projector, enabling end-to-end visual understanding and instruction following. LLaVA [5] established visual instruction tuning—aligning a vision encoder to a frozen LLM via a linear projection followed by instruction fine-tuning—as a scalable recipe for open multimodal assistants. LLaVA-OneVision [4] consolidates this line into a single open model spanning single-image, multi-image, and video tasks through a staged curriculum: stage-1 projector pre-training, stage-1.5 mid-training, and stage-2 instruction tuning. BaiZe follows this curriculum in stage (iv) (Section 5), substituting a from-scratch vision encoder (stage iii) for the frozen off-the-shelf encoder.

2.2 LLMs for Electronic Design Automation

A growing line of work applies LLM agents to chip design tasks such as register-transfer-level generation, verification, and bug triage [6, 1]. These applications motivate a cheap, always-on, agentic multimodal core that can run per-team or per-engineer; our work supplies the efficient multimodal backbone such systems require.

3 Architecture and Pipeline Overview

BaiZe is a multimodal EDA agent assembled in three stages on top of the BaiZe language backbone [7]: (i) from-scratch pre-training of a custom vision encoder, (ii) alignment of the encoder and the backbone into a multimodal large language model, and (iii) semiconductor-domain adaptation via reinforcement learning.

This section fixes the shared scaffolding—the vision encoder, the alignment projector, and the global training environment—before each stage is detailed in its own section.

The language backbone is the BaiZe $\approx$2.2B hybrid model, fixed and imported from its companion paper [7]; we do not re-train or re-describe it here, and keep it frozen throughout every stage.

3.1 Vision Encoder

The vision encoder is pre-trained from scratch in stage (i) and is selected from iso-parameter candidates at the 500–600 M scale; Section 4 reports the comparison and its outcome.

Table 1: BaiZe multimodal pipeline overview.

Stage	Task	Status	Section
(i)	Vision encoder	complete	4
(ii)	MLLM alignment	in progress	5
(iii)	Domain adapt. (RL)	CPU pilot	6

3.2 Pipeline and Global Environment

Table 1 summarises the three-stage pipeline. The vision encoder and alignment stages run on a single $8\times$H100 node (CUDA 12.8, PyTorch 2.8.0, Python 3.10); the aligned multimodal model uses a 196-token-per-image budget throughout.

4 Vision Encoder Pretraining

Stage (i) pre-trains a custom vision encoder from scratch; we summarise the outcome here and defer the design study to a companion report.

Selected encoder. We adopt **OpenVision2**, a pure-attention ViT at width-512/depth-30 (126.8M parameters), trained from scratch with a bidirectional trunk at 224px/16-px patches on a $\approx$18.5M-pair image–text corpus (CC12M + Amshaker). The objective combines a global InfoNCE contrastive term with a caption-*independent* masked-patch reconstruction term at a 1:1 weight and a 0.6 mask ratio.

Final result. On the frozen trunk with an ImageNet-1k linear probe, the encoder reaches $11.50\% \rightarrow \mathbf{19.76}\%$ top-1 from 5.1M to 55.3M samples (CC12M+Amshaker, 18.5M unique pairs), and 20.27% at 176M samples on a 58.8M-pair corpus (adding GPIC), with no saturation. This is $+12.4$ pp above a contrastive-only baseline at the same sample budget, and it is the operating point used by stage (ii). The $224/16$ resolution is retained because it is best on both loss and throughput; whether 196 visual tokens suffice for detail-dense layout and schematic views is tested in stage (ii) (Section 5).

Encoder upgrade. A second tower—ViT-L/14 (304M) trained with open_clip under the same objective but lower peak LR (5×10^{-4}) and longer warm-up (3000 steps)—reaches $\mathbf{68.236 \pm 0.018}\%$ top-1 (5-NN 58.904%) on ImageNet-1k (Protocol B, 3 seeds), versus 64.337% / 55.956% for the ViT-B/16 (86M) tower; the $+3.9$ pp gap is a joint effect of tower size, LR, and warm-up, not a capacity-only attribution.

5 MLLM Alignment

Stage (ii) connects the vision encoder (Section 4) to the language backbone via a lightweight projector, following the staged curriculum of open multimodal assistants; layout and schematic views account for 70.2% of benchmark tasks [8], so a multimodal core shares the same execution-based protocol as the text-only stage [7].

Tasks and curriculum. Three task families each admit objective ground truth: (i) *image→script* (rendered view → PyAether/SKILL script, self-labelled by execution); (ii) *script→image* (predict geometry, self-labelled); (iii) *layout VQA* (attribute questions, expert-labelled). The first family shares the assertion sandbox of the text-only stage [7], giving a common metric. We run stage-1 projector pre-training on captions, stage-1.5 mid-training on the three families, and stage-2 instruction tuning on the full mixture; ablations vary the projector type, tower freezing, and visual-token count. The $224/16$ setting (196 tokens/image) is retained; whether it transfers to detail-dense line drawings is what stage (ii) tests.

Downstream results. Table 2 reports the Stage-2 step-3500 checkpoint on 12 lmms-eval benchmarks for ViT-L/14 (304M) and ViT-B/16 (86M). The 11 accuracy benchmarks average to 26.10%

Table 2: MLLM downstream results (Stage-2 step 3500, %; internal lmms-eval, single seed, 224-px single-tile). MME† is excluded from the 11-set mean.

Benchmark	ViT-L	ViT-B	Benchmark	ViT-L	ViT-B
AI2D	60.65	60.49	CountBench	10.59	9.98
RealWorldQA	45.36	46.41	DocVQA	7.35	7.44
MMMU_val	44.00	42.67	OCRBench	1.60	1.60
BLINK	38.88	39.59	InfoVQA	16.18	16.59
MMStar	27.17	27.07	ChartQA	12.16	12.92
MMMU_Pro	23.12	22.60	**11-set avg**	**26.10**	**26.12**
MME† (perc./cog./total): 513.75/204.29/718.04				559.67/246.79/806.45	

vs. 26.12% (-0.02 pp, indistinguishable); despite the $+3.9$ pp encoder advantage (Section 4), downstream metrics are flat, suggesting the bottleneck lies not in tower capacity but in the 224-px single-tile setting, the MLLM architecture, or weak proxy-to-downstream transfer.

6 Domain Adaptation: Waveform Understanding as a Probe Skill

Stage (iii) adapts the multimodal core to semiconductor design practice. Rather than deferring this stage until a preference dataset matures, we treat *waveform understanding*—reading simulation waveforms (VCD/FSDB) to answer questions about period, duty cycle, rise time, and signal shape—as a concrete, verifiable probe skill. It is the EDA sub-task where a multimodal model is most clearly distinguished from a text-only agent [7]: an engineer *looks* at a waveform, and the relevant benchmarks down-grade precisely the visual judgements (eye-diagram aperture, edge margin, glitch detection) to text or Tcl when a vision channel is absent.

Setup. We pilot the stage entirely on CPU, consistent with the deployment constraint that proprietary data stays on-premise on the BaiZe core [7]. The base model is Qwen3.5-2B (1.88B) with LoRA [3] (r=16, ≈6.4M trainable parameters), trained in fp32 under `nice -19` with a 16-thread cap, never touching the GPUs reserved for stage (i)/(ii). A pure-Python waveform synthesizer generates sine/square/triangle/sawtooth signals with randomised amplitude, offset, frequency, phase and noise; the 16 uniform (t, v) sample points are placed directly in the text prompt so that the language model can in effect "see" the shape. All numbers below are reproducible from the logged JSON in our experiment archive (train/eval use disjoint RNG seed ranges, so the cheat rate is zero).

Reward and task redesign. An advantage-variance diagnostic on the original objective revealed the bottleneck: with a 5-bin discrete reward and group size 8, 73.0% of groups produced *identical* rewards across all rollouts, so the GRPO [9] advantage was zero and the gradient carried no signal; the zero-advantage share was *not* the "all-zero" failure mode (0.0%) but the "all-same-bin" mode (15.0%). Per-target, *period* was the only target with usable signal (std>0 in 78.3% of groups), whereas *rise_time* was 100% zero-variance and *duty_cycle* 91.7%—a capability gap, not a compute gap. We therefore redesigned on two axes: (i) a *continuous* reward $R = \exp(-\text{rel_err}/\tau)$ replacing the discrete bins, and (ii) *task redefinition* from exact-number regression to a discriminative intermediate task (waveform *type classification*, then *frequency comparison*), which is decidable, informative for downstream use, and far more learnable for a 2B adapter. The adaptation methods are rejection-sampling fine-tuning (RFT, in the STaR/ReST family [10, 2]) and outcome-only GRPO with a KL-free group baseline.

Protocol and falsification. Every arm is pre-registered before measurement: four conditions are evaluated on a fixed held-out set with ≥ 3 random seeds. **A** = fresh LoRA baseline; **B** = A after SFT; **C** = B after the adaptation step (RFT or GRPO); **D** = control (random-filter RFT or random-reward GRPO). Success requires $C{>}B$, $C{>}D$, a McNemar test on C vs B with $p{<}0.05$, and a bootstrap 95% CI on $C{-}B$ that excludes zero. Negative results are reported as-is; a CPU pilot is never extrapolated to a GPU claim.

Results. Table 3 summarises seven consecutive design iterations. The single statistically significant positive is the type-classification task with RFT (*v5*): accuracy climbs $A{=}16.0\% \rightarrow B{=}37.0\% \rightarrow C{=}53.0\%$, with McNemar C-vs-B $p{=}0.0009$ and bootstrap CI $[+7\%, +25\%]$, while the random-

Table 3: Stage (iii) CPU pilot on Qwen3.5-2B + LoRA. Accuracy (%) per pre-registered arm; NS = McNemar $p \geq 0.05$. All runs: train/eval disjoint seeds $\Rightarrow$ cheat rate $= 0$. "–" = not applicable to that task.

Iter	Task (method)	A	B	C	D (ctrl)	McNemar $C vs. B$	Verdict
v5	Type cls., 4-way (SFT→RFT)	16.0	37.0	**53.0**	32.0	p=0.0009	**Significant**
v6	Type cls. (SFT→GRPO)	–	35.0	40.0	–	p=0.267	NS
v7	Type cls. (GRPO+process)	–	∼40	49.3±6.4	–	NS	6/6 crit., NS test
v8	Freq. compare (SFT→GRPO)	0.0	45.0	52.5±6.1	45.0	p=0.031 (s0)	Mixed
v9	Freq. compare (GRPO)	0.0	49.5	49.5	–	p=1.0	No change
v10	Freq. compare (GRPO)	46.2	54.9	51.6±3.1	53.8	p=1.0	NS
v11	Regen. vs. direct	22.2	–	20.8 / 13.2*	–	–	0/6, negative

*v11 C columns are *direct / augment / synthesis* arms; generative augmentation is worse than direct.

filter control D=32.0% falls *below* B—the gain is driven by the correctness filter, not by data volume. The improvement concentrates on the visually distinctive *square* class (84%); *triangle*/*sawtooth* remain near zero, a persistent capability gap. Switching the adapter from RFT to GRPO on the same task (*v6*) drops to 40% (McNemar p=0.267, NS); adding a process-reward term (*v7*) reaches $49.3\% \pm 6.4\%$ and passes 6/6 pre-registered criteria yet fails the McNemar test because the seed-0 baseline runs high. Redefining the task to *frequency comparison* (*v8*) yields a mixed $52.5\% \pm 6.1\%$ (significant for one seed only), and three further iterations (*v9–v11*)—including a generative-augmentation route—produce no advantage over the direct arm (*v11*: direct 22.2% vs. augment 20.8% vs. synthesis 13.2%, $0/6$ criteria). We read this honestly: *one* task–method pairing gives a significant, controlled gain, and it does *not* transfer to the harder regression or frequency tasks within the CPU budget.

Where vision would actually help. To bound the value of a true multimodal (image) channel, we audited the waveview-related cases of an EDA-flow benchmark [8]: of 37 waveform-related cases (30 failed), 4 (α=13.3%, optimistic $\alpha+\beta \approx 36\%$) fail *directly because* the model cannot see the waveform shape—eye-diagram aperture, rise-edge margin, root-cause diff, functional inference from a plot. This audit was conducted under a benchmark version that *downgraded* the visual judgement of 7 evaluated cases to text/Tcl (all passing); downgrading the very capability under test changes the ruler, so that version has since been corrected—need-to-see-image cases are retained in original form and a system without multimodal ability is now *expected* to fail them. The α/β cases therefore become the measurement substrate for multimodal gain, defined as the *baseline-fail* $\rightarrow$ *with-vision-pass* delta, which the cross-line re-evaluation below targets. The implication for stage (iii) is that the image-channel ceiling must be re-measured under the corrected benchmark rather than read off the downgraded numbers, and that task/reward design (above) together with a waveform-VQA set built from the α/β cases are the high-ROI levers.

Honest status. Within a CPU-only budget, stage (iii) delivers one controlled, significant result (type classification, +16 pp over SFT, p=0.0009, cheat rate 0) and a sequence of non-significant or negative results that we report rather than suppress; the binding constraint is reward/task definition, not compute, so waiting for GPUs would not have changed the qualitative outcome. Two items remain open and are stated as such rather than estimated: (i) scaling the significant pairing to GPU-scale GRPO, and (ii) the cross-line re-evaluation that wraps Qwen3.5-2B as a VQA tool and re-runs the α/β waveview cases against a text-only baseline—the dependency chain whose first link is the audit above.

7 Conclusion

We built the multimodal core of BaiZe on the frozen BaiZe language backbone [7]: a vision encoder pre-trained from scratch, aligned to the backbone through a lightweight projector, and adapted to the semiconductor domain via preference-based reinforcement learning.

Through a controlled encoder comparison, we found that a larger ViT-L/14 tower (68.2% linear-probe, +3.9 pp over ViT-B/16) does *not* translate its advantage downstream: the aligned MLLM averages 26.1% over 11 benchmarks, indistinguishable from the smaller tower. We attribute the gap to the 224-px single-tile setting and the alignment architecture rather than to tower capacity.

Stages (ii) and (iii) are evaluated across 12 general multimodal benchmarks, 4 domain benchmarks, and the multimodal subset of an EDA flow benchmark. Stage (iii) is reported here as a CPU-only pilot (Section 6): it yields one controlled, statistically significant gain on a waveform type-classification probe ($+16$ pp over SFT, $p{=}0.0009$, cheat rate 0) alongside a sequence of honestly reported non-significant or negative iterations, and an audit that bounds the ceiling of an image channel on the current benchmark. Scaling the significant pairing to GPU-scale RL, and the cross-line VQA-tool re-evaluation of the waveview cases, remain open and are stated as such rather than estimated.

The immediate next step is closing the encoder-to-downstream transfer gap—through higher-resolution multi-tile input and domain-specific alignment—toward a deployable multimodal core for visual EDA tasks.

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